ELEC70048
EE9-CS5-1
EE9-SC3
EE9-AML2-01

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE PART IV: MEng and ACGI

PROBABILITY AND STOCHASTIC PROCESSES

Tuesday, 29 April 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Information for students

Each of the four questions has 25 marks.

The Questions

1. Random variables.

a) Let the joint probability density function of X and Y be given by

$$f_{XY}(x,y) = \begin{cases} e^{-x} & 0 < y \leq x \leq \infty \\ 0 & \text{otherwise} \end{cases}$$

Define $Z = X + Y, W = X - Y$.

i) Find the joint probability density function of Z and W. [5]

ii) Derive the probability density function of Z. [5]

b) A random variable X has the probability density function:

$$f_X(x) = \begin{cases} kx^2, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

Find the value of k and compute $E[X]$. [10]

c) A fair die is rolled twice. What is the probability that the sum of the outcomes is 7?

[5]

2. Estimation.

A random variable X has a Poisson distribution with parameter λ

$$P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \quad k = 0,1,2,\ldots$$

We wish to estimate λ from n i.i.d. samples of X.

a) Compute the Fisher information

$$I(\lambda) = -E\left[\frac{\partial^2}{\partial\lambda^2}\ln f_X(x_1,\cdots,x_n;\lambda)\right].$$

[10]

b) Compute the Cramer-Rao lower bound (CRLB).

[5]

c) Consider the sample mean

$$\hat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

i) Show whether it is an unbiased estimator of λ.

ii) Show whether it is a consistent estimator of λ.

iii) Show whether it is an efficient estimator of λ.

[10]

3. Random processes.

a) Consider the matched filter shown in the following figure:

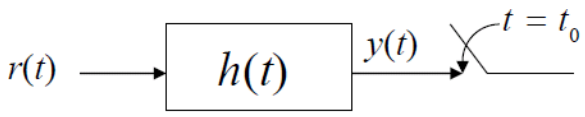


Fig. 3.1. Matched filter.

Let $r(t)$ represent a deterministic signal $s(t)$ corrupted by noise

$$r(t) = s(t) + w(t), \qquad 0 < t < t_0 = 1$$

where $w(t)$ is additive white noise with power spectral density $N_0 = 1$. Let $s(t)$ be a sinusoidal signal

$$s(t) = \begin{cases} \frac{1}{2}\cos(\pi t), & 0 < t < 1 \\ 0, & \text{otherwise} \end{cases}$$

i) Find the expression of the optimum matched filter $h(t)$. [5]

ii) Find the maximum output signal-to-noise ratio (SNR). [5]

b) Let $Z_1, Z_2, \ldots$ be uncorrelated random variables with zero mean and unit variance. You may assume the period $T = 1$ without loss of generality.

i) Define the moving average process $X_n = Z_n + \alpha Z_{n-1}$ where α is a constant. Find the power spectrum density of $\{X_n\}$. [6]

ii) More generally, let $Y_n = \sum_{i=0}^{r} \alpha_i Z_{n-i}$ where $\alpha_0 = 1$ and $\alpha_1, \ldots, \alpha_r$ are constants. Find the power spectrum density of $\{Y_n\}$. [9]

4. Markov chains.

a) A Markov chain has three states: A, B, and C, with transition probabilities:

	A	B	C
A	0.4	0.5	0.1
B	0.2	0.3	0.5
C	0.3	0.3	0.4

i) Find the steady-state distribution of the chain. [5]

ii) If the system starts in state A, what is the probability that it is in state B after two steps? [5]

b) In an old house, white mice and black mice arrive as independent Poisson processes with intensity 0.2 and 0.25, i.e., at a rate of 1 white mouse/5 minutes and at a rate of 1 black mouse/4 minutes, respectively.

i) What is the probability that the first mouse is black? [5]

ii) What is the probability that the first mouse is black while the second is white? [5]

iii) Two types of cats (the numbers of cats are unlimited) are waiting in the house so that each mouse is caught randomly with equal probability. One type catches a mouse every 10 minutes, and the other type catches a mouse every 20 minutes. Find the average number of mice in the house. [5]